Electric Circuits

ELEC 221 – Electrical Circuits

Alireza Bakhshai, PhD

Learning Outcomes

By the end of this lecture, you will learn more about:

- Inductors.
- Power and energy in inductors.
- Initial conditions in inductors.
- Capacitors.
- Power and energy in capacitors.
- Initial conditions in capacitors.

Chapter 6: Introduction

Last 2 **ideal** elements encountered in this course are introduced:

Capacitors (C) and **Inductors (L)**: passive elements.

- C and L are elements with "**memory**".
- R does not have memory. Why?
- Ohm law governs the terminal behavior of R: voltage at time *t* is only a function of current **at the same** time *t*.
- Memory ⇨ nifty and funky dynamics to be expected!

Introduction (cont.)

- What happens to chapter 4 laws, theorems, and techniques?
  - **They are all valid**. Reminder: KVL, KCL, and conservation of power/energy (and all based on these) stem from fundamental laws of physics.

- All we need to add: C and L's terminal behavior: how v and i are related.

- Like other elements, L and C are easier to work with in terms of circuit variables rather than electromagnetic field variables.

What is an Inductor?

- L: an element which **opposes changes in electric current**!
- Flux is a function of current: $\Phi = f(I;t)$

  Magnetic flux (Weber)= magnetic field (B, Tesla) x area (A)

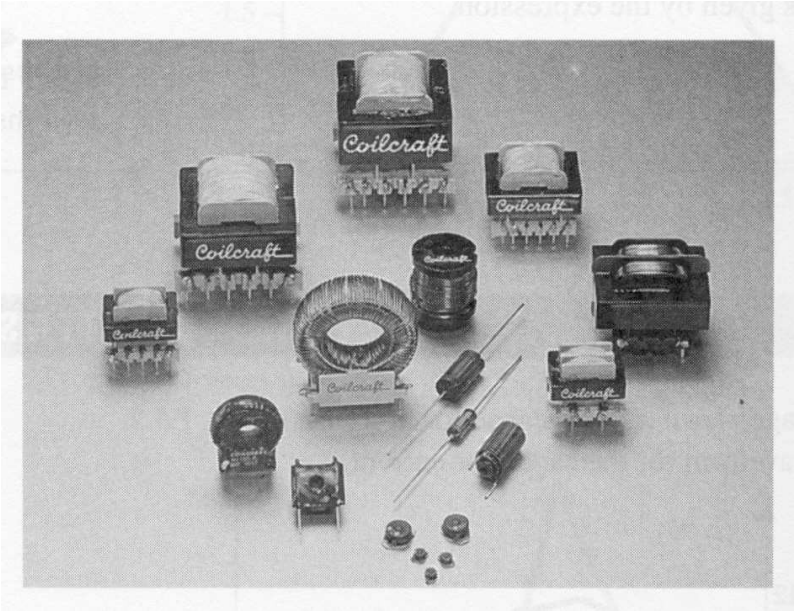
  - Linear inductors: flux is a linear function of current: $\Phi = L(t).I$
  - Time-invariant (TI): flux is a time-invariant function of current: $\Phi = f(I)$
- In 221: we focus on **LTI inductors**: $\Phi = L.I$

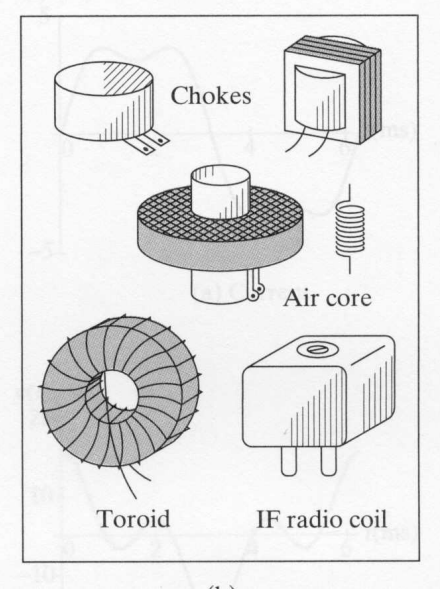
What is an Inductor? (cont.)

- Inductors are composed of a coil of wire wrapped around a **core** which can be:
  - Magnetic: iron or ferrite (higher frequencies), e.g., ferric oxide (high magnetic permeability and high electric resistance).
  - Nonmagnetic: air or any other low permeability material (radio and TV circuitry).

Inductors (cont.)

Inductors:





Solenoid

Inductors (cont.)

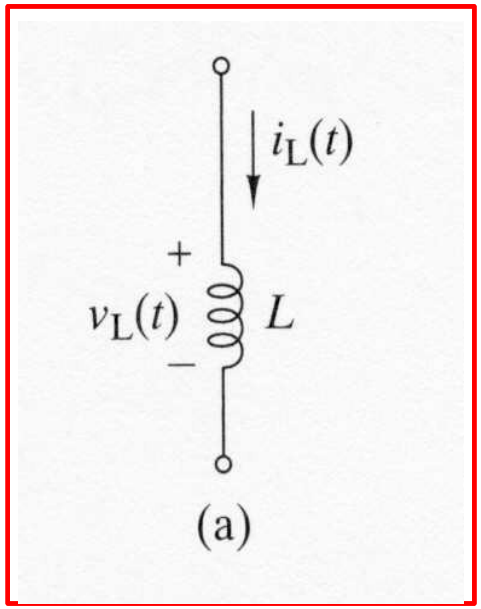
Charge in motion or **current** ⇨ **magnetic field** around the wire

Varying current ⇨ **varying** magnetic field

Varying magnetic field ⇨ a **voltage** in the conductor

$$v_L(t) = \frac{d\varphi}{dt} = L\frac{di_L(t)}{dt}$$

Inductance in Henry (H)



1- Inductors are electric elements that oppose any change in current.

2- Current cannot change abruptly. Otherwise, voltage will be **infinite**!!!

3- If $i_L = constant$ ⇨ $v_L = 0$. That is, for DC current the inductor = **short circuit**.

Inductors (cont.)

Of course, the infinite voltage statement is only true in theory. What does happen in practice?

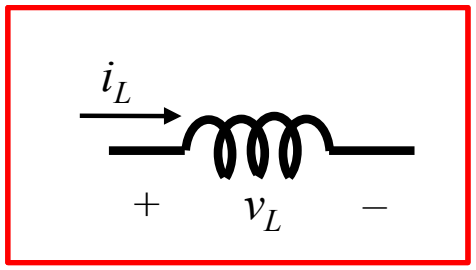
- **Arcing:** current will flow in the air in the switch across the inductor due to high voltage generated.
- Arcing can damage the circuit and elements: should be avoided.

Inductors (cont.)

Standard inductor values

nH	nH	nH	µH	µH	µH	mH	mH	mH
1	10	100	1.0	10	100	1.0	10	100
1.2	12	120	1.2	12	120	1.2	12	
1.5	15	150	1.5	15	150	1.5	15	
1.8	18	180	1.8	18	180	1.8	18	
2	20	200	2.0	20	200	2.0	20	
2.2	22	220	2.2	22	200	2.2	22	
2.7	27	270	2.7	27	270	2.7	27	
3	33	330	3.3	33	330	3.3	33	
4	39	390	3.9	39	390	3.9	39	
5	47	470	4.7	47	470	4.7	47	
6	51	510	5.1	51	510	5.1	51	
7	56	560	5.6	56	560	5.6	56	
8	68	680	6.8	68	680	6.8	68	
9	82	820	8.2	82	820	8.2	82	

Inductors: Current and Voltage Relationship

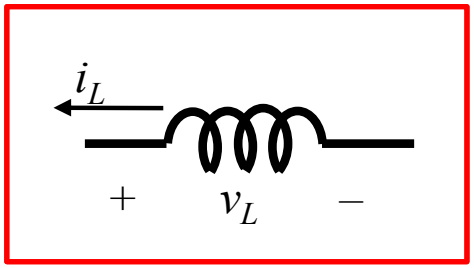


$$v_L = L\frac{di_L}{dt} \Rightarrow di_L = \frac{1}{L}v_L dt \Rightarrow \int_{i_L(t_0)}^{i_L(t)} dx = \frac{1}{L}\int_{t_0}^{t} v_L d\tau$$

$$\Rightarrow i_L(t) - i_L(t_0) = \frac{1}{L}\int_{t_0}^{t} v_L d\tau \quad \boxed{\Rightarrow i_L(t) = \frac{1}{L}\int_{t_0}^{t} v_L d\tau + i_L(t_0)}$$

Do you see the memory of the element?

Inductors: Current and Voltage Relationship

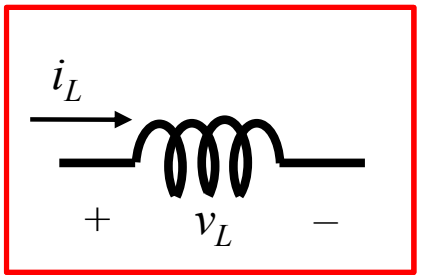


$$v_L = -L\frac{di_L}{dt} \Rightarrow di_L = -\frac{1}{L}v_L dt \Rightarrow \int_{i_L(t_0)}^{i_L(t)} dx = -\frac{1}{L}\int_{t_0}^{t} v_L d\tau$$

$$\Rightarrow i_L(t) - i_L(t_0) = -\frac{1}{L}\int_{t_0}^{t} v_L d\tau \quad \Rightarrow i_L(t) = -\frac{1}{L}\int_{t_0}^{t} v_L d\tau + i_L(t_0)$$

Do you see the memory of the element?

Inductors: Power and Energy



$$v_L = L\frac{di_L}{dt}$$

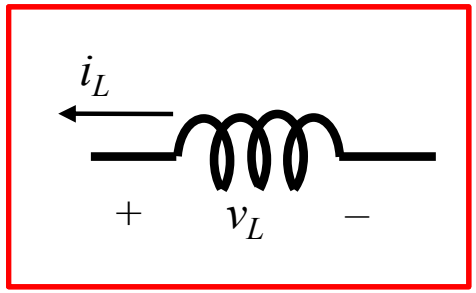
$$\left.\begin{aligned} &\boxed{p_L = v_L i_L = L i_L \frac{di_L}{dt}} \\ &p_L = \frac{dw_L}{dt} \Rightarrow dw_L = p_L dt \end{aligned}\right\} \Rightarrow dw_L = L i_L di_L \Rightarrow \int_{w_L(t_0)}^{w_L(t)} dx = L\int_{i_L(t_0)}^{i_L(t)} y dy$$

$$\Rightarrow w_L(t) - w_L(t_0) = \frac{1}{2}Li_L^2(t) - \frac{1}{2}Li_L^2(t_0)$$

$$\boxed{\Rightarrow w_L(t) = \frac{1}{2}Li_L^2(t)}$$

Energy stored in the inductor magnetic field at time *t*.

Inductors: Power and Energy (cont.)



$$v_L = -L\frac{di_L}{dt}$$

$$p_L = -v_L i_L = +Li_L\frac{di_L}{dt}$$

$$p_L = \frac{dw_L}{dt} \Rightarrow dw_L = p_L dt$$

$$\Rightarrow w_L(t) = \frac{1}{2}Li_L^2(t) > 0$$

Energy stored in the inductor magnetic field at time t

$p_L = Li_L(t)\dfrac{di_L(t)}{d(t)}$ can be positive or negative, but $w_L = \dfrac{1}{2}Li_L{}^2(t)$ is always positive

($w_L(t) > 0$, passive element).

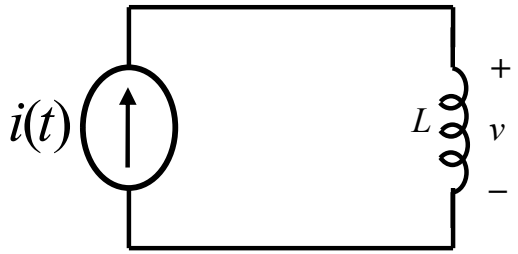
An inductor never dissipates or generates energy.

It can only store and release the stored energy.

Example: Inductors

In the following simple circuit, *L= 100 mH, and* :

$$\begin{cases} i(t) = 10te^{-5t}, t \geq 0 \\ i(t) = 0, t < 0 \end{cases}$$



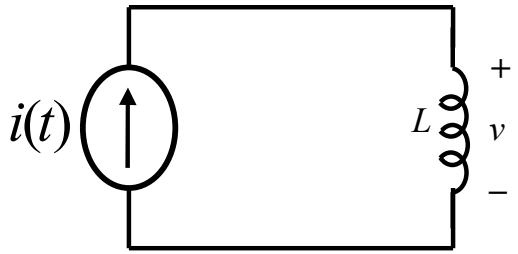
- Sketch the current waveform.
- Find and sketch the voltage across the inductor.
- Are ***v*** and ***i*** maximum at the same time? Discuss.
- Of ***v*** and ***i*** across L which one has an abrupt change?
- Find an plot power an energy of L.
- When does L store energy? When does it deliver energy?

Do examples 6.1, 6.2, and 6.3 of the text.

Example: Inductors

In the following simple circuit, *L= 100 mH, and* :

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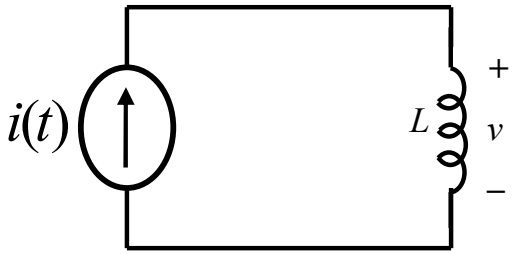


- Sketch the current waveform.

Example: Inductors

In the following simple circuit, *L= 100 mH, and* :

$$\begin{cases} i(t) = 10te^{-5t}, t \geq 0 \\ i(t) = 0, t < 0 \end{cases}$$



- Sketch the current waveform.
- Find and sketch the voltage across the inductor.
- Are ***v*** and ***i*** maximum at the same time? Discuss.
- Of ***v*** and ***i*** across L which one has an abrupt change?
- Find an plot power an energy of L.
- When does L store energy? When does it deliver energy?

Do examples 6.1, 6.2, and 6.3 of the text.

Example: Inductors (cont.)

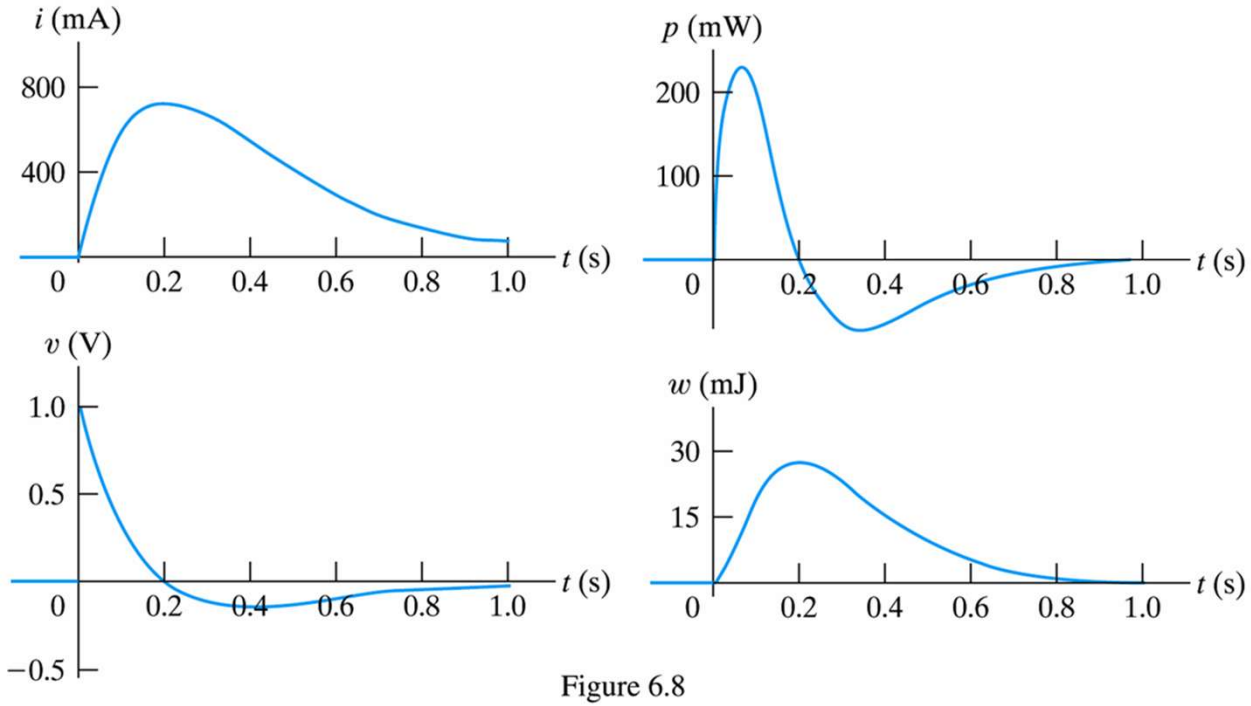


Figure 6.8

The variables i, v, p, and w versus t for Example 6.1.

What is an Capacitor?

- C: an element which **opposes changes in electric voltage**!

- Charge is a function of voltage: $Q = f(v;t)$
  - Linear capacitors: charge is a linear function of voltage: $Q = C(t).v$
  - Time-invariant (TI): charge is a time-invariant function of voltage: $Q = f(v)$

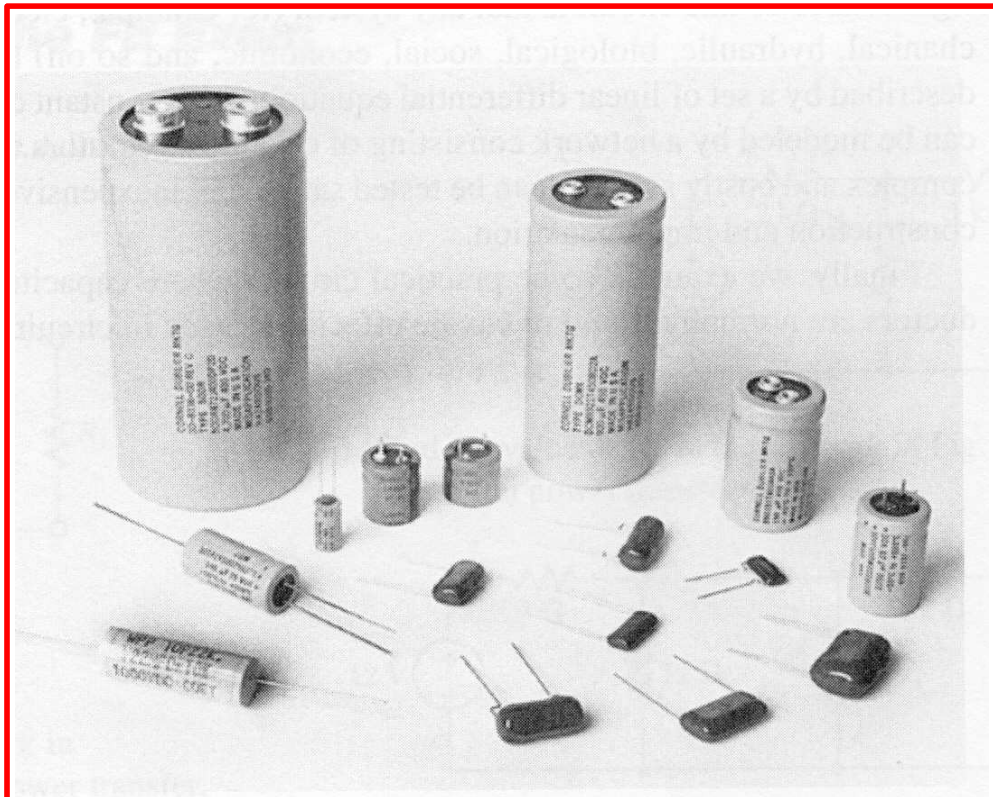
- In 221: we focus on **LTI capacitors**: $\boxed{Q = C.v}$

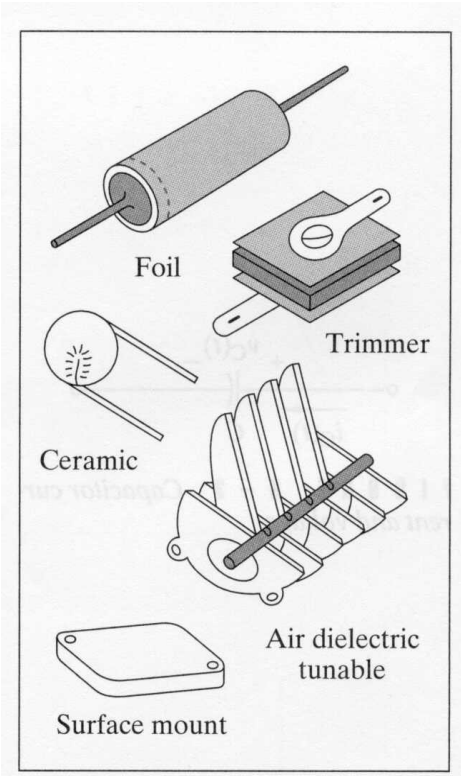
What is a Capacitor? (cont.)

- Capacitors are made of two conductors separated by
  - ✓ An insulator, or,
  - ✓ A dielectric material (high resistance but good supporter of electric fields: better choice for higher frequencies).
  - ✓ E.g., Mica, glass, ceramic, Aluminum or Tantalum electrolytic (high capacitance and small volume).

Capacitors

- **Capacitors:**





Capacitors (cont.)

Voltage across capacitor terminals ⇨ separation and transfer of opposite charges to opposite terminals via the **external** circuit (**charge displacement**).

Charges cannot cross the dielectric: no conduction current (indistinguishable at the terminals).

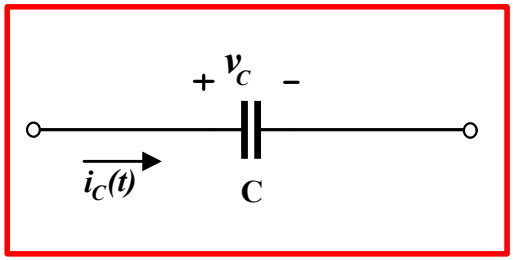
Opposite charges on two plates ⇨ **Electric field**.

Varying voltage ⇨ charge displacement or **displacement current** .

Capacitors (cont.)

$$i_c(t) = \frac{dQ}{dt} = C\frac{dv_c(t)}{dt}$$

Capacitance in Farad (F)



1- Capacitors are electric elements that **oppose any change in voltage.**

2- Voltage **cannot change abruptly**. Otherwise, current will be infinite.

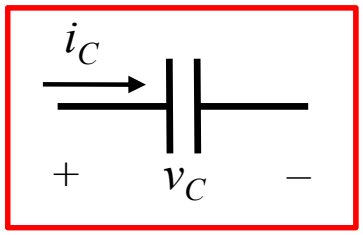
3- If $v_c = constant$ ⇨ $i_c = 0$: For DC voltage, Capacitor acts like open circuit.

Capacitors (cont.)

Standard capacitor values

pF	pF	pF	pF	μF	μF	μF	μF	μF	μF	μF
1	10	100	1000	0.010	0.10	1.0	10	100	1000	10,000
	12	120	1200	0.012	0.12	1.2	12	120	1200	12,000
1.5	15	150	1500	0.015	0.15	1.5	15	150	1500	15,000
	18	180	1800	0.018	0.18	1.8	18	180	1800	18,000
2	20	200	2000	0.020	0.20	2.0	20	200	2000	20,000
	22	220	2200	0.022	0.20	2.2	22	220	2200	22,000
	27	270	2700	0.027	0.27	2.7	27	270	2700	27,000
3	33	330	3300	0.033	0.33	3.3	33	330	3300	33,000
4	39	390	3900	0.039	0.39	3.9	39	390	3900	39,000
5	47	470	4700	0.047	0.47	4.7	47	470	4700	47,000
6	51	510	5100	0.051	0.51	5.1	51	510	5100	51,000
7	56	560	5600	0.056	0.56	5.6	56	560	5600	56,000
8	68	680	6800	0.068	0.68	6.8	68	680	6800	68,000
9	82	820	8200	0.082	0.82	8.2	82	820	8200	82,000

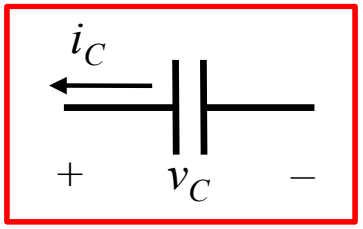
Capacitors: Voltage and Current Relationship



$$i_C = C\frac{dv_C}{dt} \Rightarrow dv_C = \frac{1}{C}i_C dt \Rightarrow \int_{v_C(t_0)}^{v_C(t)} dx = \frac{1}{C}\int_{t_0}^{t} i_C d\tau$$

$$\Rightarrow v_C(t) - v_C(t_0) = \frac{1}{C}\int_{t_0}^{t} i_C d\tau \qquad \boxed{\Rightarrow v_C(t) = \frac{1}{C}\int_{t_0}^{t} i_C d\tau + v_C(t_0)}$$

PSC used here!

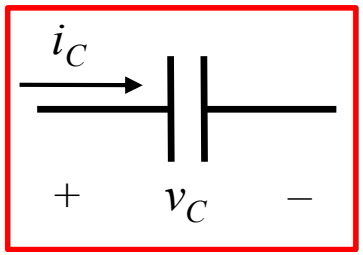


$$i_C = -C\frac{dv_C}{dt} \Rightarrow dv_C = -\frac{1}{C}i_C dt \Rightarrow \int_{v_C(t_0)}^{v_C(t)} dx = -\frac{1}{C}\int_{t_0}^{t} i_C d\tau$$

$$\Rightarrow v_C(t) - v_C(t_0) = -\frac{1}{C}\int_{t_0}^{t} i_C d\tau \qquad \boxed{\Rightarrow v_C(t) = -\frac{1}{C}\int_{t_0}^{t} i_C d\tau + v_C(t_0)}$$

Do you see the memory of the element?

Capacitors: Power and Energy



$$i_C = C\frac{dv_C}{dt}$$

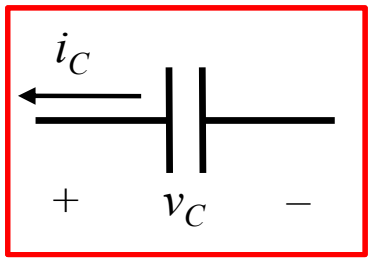
$$\left.\begin{aligned} &p_C = v_C i_C = Cv_C\frac{dv_C}{dt} \\ &p_C = \frac{dw_C}{dt} \Rightarrow dw_C = p_C dt \end{aligned}\right\} \Rightarrow dw_C = Cv_C dv_C$$

$$\Rightarrow \int_{w_C(t_0)}^{w_C(t)} dx = C\int_{v_C(t_0)}^{v_C(t)} y\,dy \quad \Rightarrow w_C(t) - w_C(t_0) = \frac{1}{2}Cv_C^2(t) - \frac{1}{2}Cv_C^2(t_0)$$

$$\Rightarrow w_C(t) = \frac{1}{2}Cv_C^2(t)$$

Energy stored in the capacitor electric field at time *t*.

Capacitors: Power and Energy (cont.)



$$i_C = -C\frac{dv_C}{dt}$$

$$p_C = -v_C i_C = +Cv_C\frac{dv_C}{dt}$$

$$p_c = \frac{dw_C}{dt} \Rightarrow dw_C = p_C dt$$

$$\Rightarrow w_C(t) = \frac{1}{2}Cv_C^2(t) > 0$$

Energy stored in the capacitor electric field at time *t*.

- $p_C = Cv_C(t)\frac{dv_C(t)}{dt}$ can be positive or negative, but $w_C = \frac{1}{2}Cv_C^2(t)$ is always positive ($w_C(t) > 0$, passive element).
- A capacitor **never dissipates or generates** energy.
  It can **store and release** the stored energy.

Initial Conditions

As L and C are elements with memory the voltage and current at a given time depends on those of past:

$$i_L(t) = \frac{1}{L}\int_{t_0}^{t} v_L d\tau + i_L(t_0)$$

For $i_L(t)$: we need $v_L(t)$ in the whole Interval $[t_0,t]$ and $i_L(t_0)$.

$$v_C(t) = \frac{1}{C}\int_{t_0}^{t} i_C d\tau + v_C(t_0)$$

For $v_C(t)$: we need $i_C(t)$ in the whole Interval $[t_0,t]$ and $v_C(t_0)$.

The latter is referred to as the **initial condition** of the elements:

Inductor: needs L and its **initial current** to be fully characterized.

Capacitor: needs C and its **initial voltage** to be fully characterized.

Duality

You might have noticed the **perfect duality** between L and C:

inductor ←→capacitor

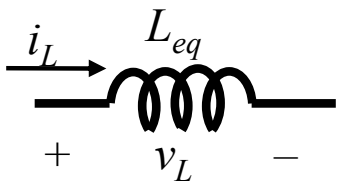
L, inductance ←→ C, capacitance

voltage ←→ current

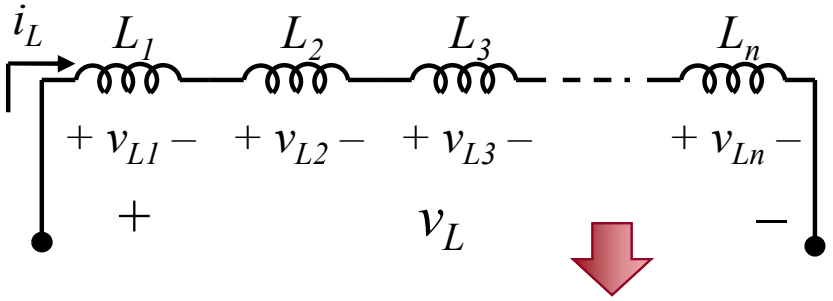
current ←→ voltage

magnetic flux ←→ charge

Inductors: Series Configuration



$$v_L = L_{eq}\frac{di_L}{dt}$$



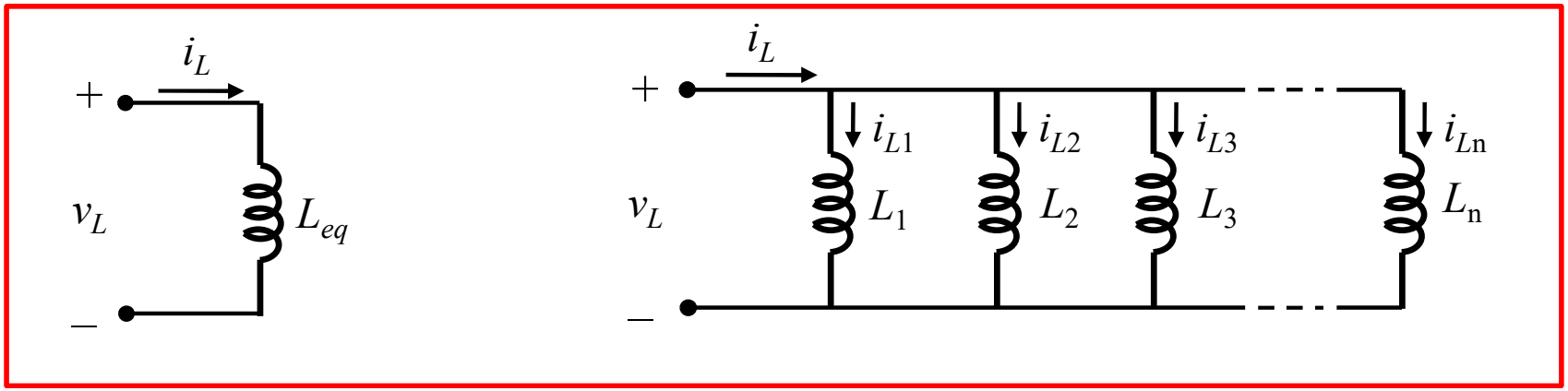
$$\begin{aligned} v_L &= v_{L1} + v_{L2} + v_{L3} + \ldots + v_{Ln} \\ &= L_1\frac{di_L}{dt} + L_2\frac{di_L}{dt} + L_3\frac{di_L}{dt} + \ldots + L_n\frac{di_L}{dt} \\ &= \left(L_1 + L_2 + L_3 + \ldots + L_n\right)\frac{di_L}{dt} \end{aligned}$$

$$\Rightarrow L_{eq} = L_1 + L_2 + L_3 + \ldots + L_n$$

As the inductors are in series, they all must have the same initial condition to be put in series validly. Why?

$$i_L(t_0) = i_{L1}(t_0) = i_{L2}(t_0) = \ldots = i_{Ln}(t_0)$$

Inductors: Parallel Configuration



$$i_L = \frac{1}{L_{eq}} \int_{t_0}^{t} v_L d\tau + i_L(t_0)$$

$$v_L = v_{L1} = v_{L2} = v_{L3} = \ldots = v_{Ln}$$

$$i_L = i_{L1} + i_{L2} + i_{L3} + \ldots + i_{Ln} \quad \text{(KCL)}$$

$$\Rightarrow i_L(t_0) = i_{L1}(t_0) + i_{L2}(t_0) + i_{L3}(t_0) + \ldots + i_{Ln}(t_0)$$

Inductors: Parallel Configuration

$$\Rightarrow i_L(t_0) = i_{L1}(t_0) + i_{L2}(t_0) + i_{L3}(t_0) + \ldots + i_{Ln}(t_0)$$

$$i_L(t) = \left[\frac{1}{L_1}\int_{t_0}^{t} v_L d\tau + i_{L1}(t_0)\right] + \left[\frac{1}{L_2}\int_{t_0}^{t} v_L d\tau + i_{L2}(t_0)\right] + \ldots + \left[\frac{1}{L_n}\int_{t_0}^{t} v_L d\tau + i_{Ln}(t_0)\right]$$

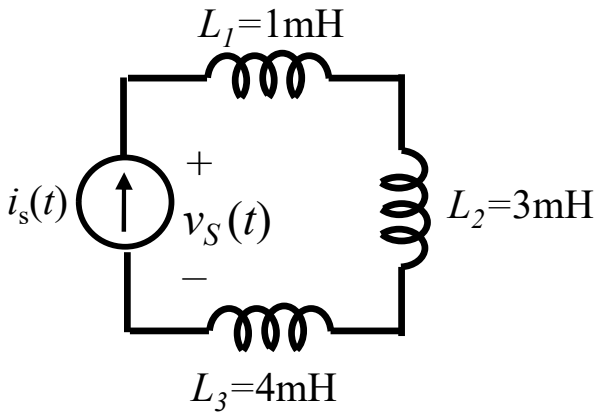
$$= \left(\frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_n}\right)\int_{t_0}^{t} v_L d\tau + \left[i_{L1}(t_0) + i_{L2}(t_0) + \ldots + i_{Ln}(t_0)\right]$$

$$= \frac{1}{L_{eq}}\int_{t_0}^{t} v_L d\tau + i_L(t_0)$$

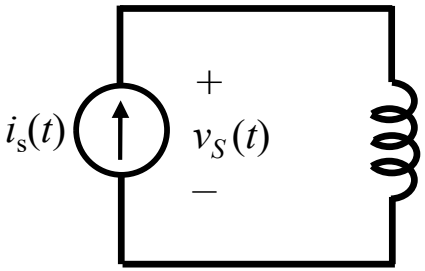
$$\Rightarrow \frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_n}$$

Series Inductors

Example: Find $v_S(t)$.



$$i_S(t) = \begin{cases} \sin t & t \geq 0 \\ 0 & t < 0 \end{cases}$$



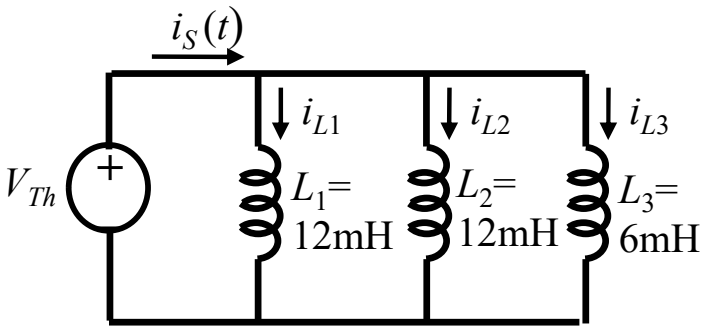
$L_{eq} = L_1 + L_2 + L_3$
$= 8\text{mH}$

$$v = L\frac{di}{dt}$$

$$v_S(t) = \begin{cases} L_{eq}\dfrac{di_S(t)}{dt} = 8\cos t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Parallel Inductors

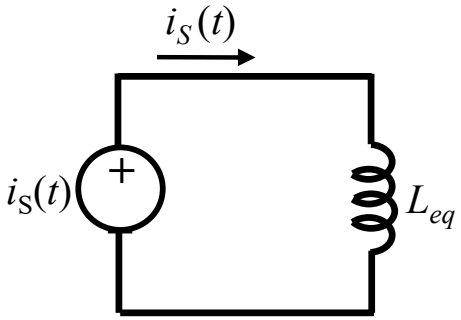
Example: Find $i_S(t)$.



$$v_S(t) = 9\cos t \qquad t \geq 0$$

$$i_{L1}(0) = i_{L2}(0) = 2\text{ mA}$$

$$i_{L3}(0) = 4\text{ mA}$$



$$\boxed{i(t) = \frac{1}{L}\int_{t_0}^{t} v\,d\tau + i(t_0)}$$

$$i_S(t) = \frac{1}{L_{eq}}\int_0^t v_S\,d\tau + i_S(0)$$

$$i_S(0) = i_1(0) + i_2(0) + i_3(0) = 8\text{mA}$$

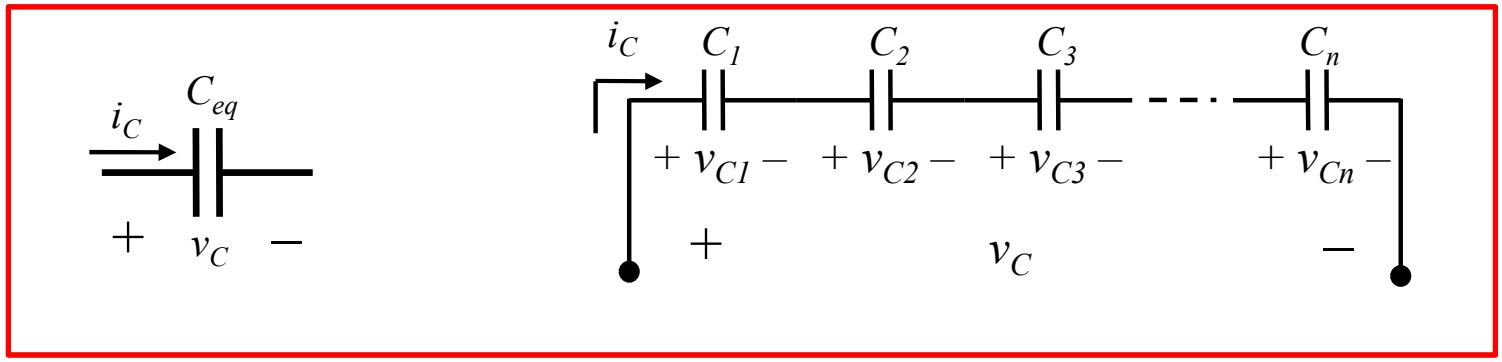
$$L_{eq} = L_1 \parallel L_2 \parallel L_3 = 12 \parallel 12 \parallel 6$$

$$= (12 \parallel 12) \parallel 6 = 6 \parallel 6 = 3\text{mH}$$

$$i_S(t) = \frac{1}{3}\int_0^t 9\cos\tau d\tau + 8 \quad = 3\int_0^t \cos\tau d\tau + 8$$

$$\boxed{\Rightarrow i_S(t) = 3\sin t + 8}$$

Capacitors: Series Configuration



$$v_C = \frac{1}{C_{eq}} \int_{t_0}^{t} i_C d\tau + v_C(t_0)$$

$$i_C = i_{C1} = i_{C2} = i_{C3} = \ldots = i_{Cn}$$

$$v_C = v_{C1} + v_{C2} + v_{C3} + \ldots + v_{Cn} \quad \text{(KCL)}$$

$$\Rightarrow v_C(t_0) = v_{C1}(t_0) + v_{C2}(t_0) + v_{C3}(t_0) + \ldots + v_{Cn}(t_0)$$

Capacitors: Series Configuration

$$\Rightarrow v_C(t_0) = v_{C1}(t_0) + v_{C2}(t_0) + v_{C3}(t_0) + \ldots + v_{Cn}(t_0)$$

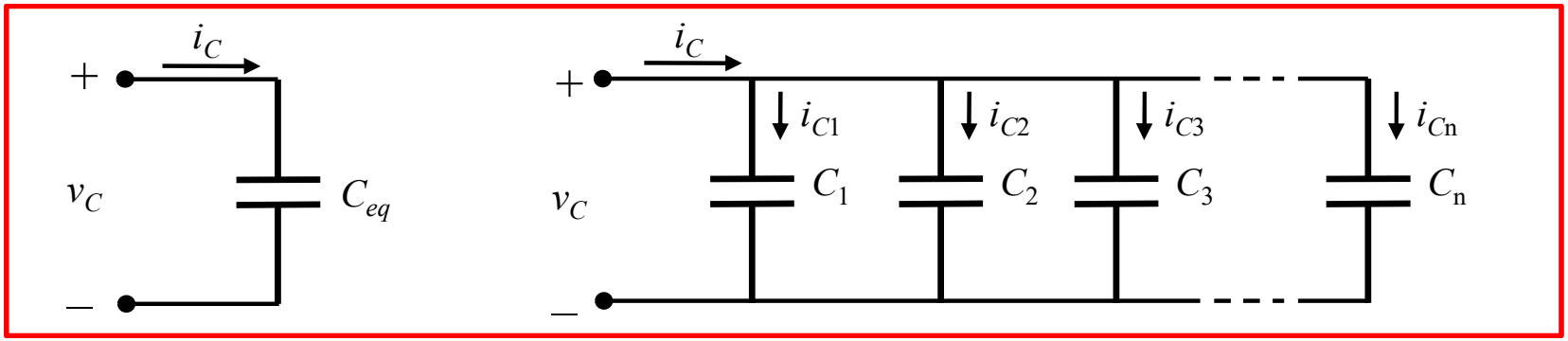
$$v_C(t) = \left[\frac{1}{C_1}\int_{t_0}^{t} i_C d\tau + v_{C1}(t_0)\right] + \left[\frac{1}{C_2}\int_{t_0}^{t} i_C d\tau + v_{C2}(t_0)\right] + \ldots + \left[\frac{1}{C_n}\int_{t_0}^{t} i_C d\tau + v_{Cn}(t_0)\right]$$

$$= \left(\frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_n}\right)\int_{t_0}^{t} i_C d\tau + \left[v_{C1}(t_0) + v_{C2}(t_0) + \ldots + v_{Cn}(t_0)\right]$$

$$= \frac{1}{C_{eq}}\int_{t_0}^{t} i_C d\tau + v_C(t_0)$$

$$\Rightarrow \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_n}$$

Capacitors: Parallel Configuration



$$i_C = C_{eq}\frac{dv_C}{dt}$$

$$i_C = i_{C1} + i_{C2} + i_{C3} + \ldots + i_{Cn} \quad \text{(KCL)}$$

$$= C_1\frac{dv_C}{dt} + C_2\frac{dv_C}{dt} + C_3\frac{dv_C}{dt} + \ldots + C_n\frac{dv_C}{dt}$$

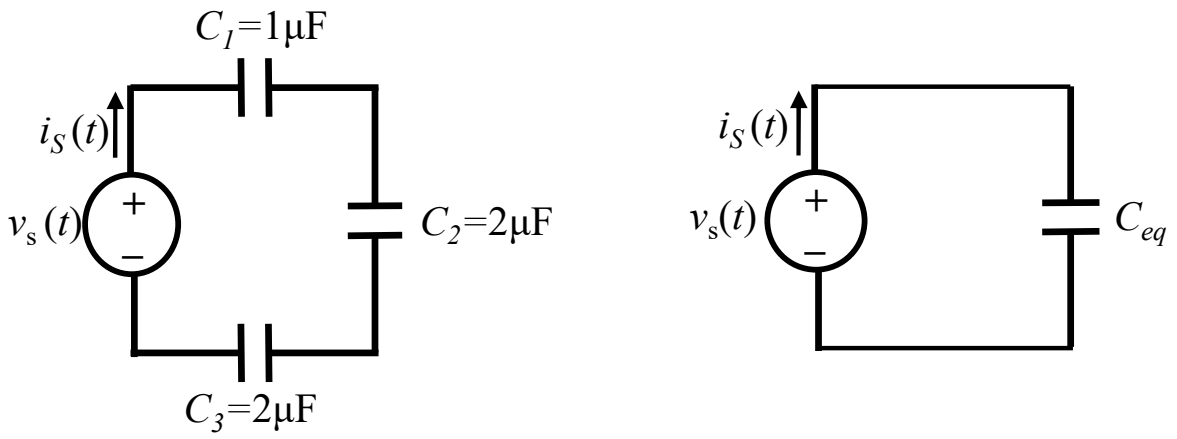
$$= \left(C_1 + C_2 + C_3 + \ldots + C_n\right)\frac{dv_C}{dt}$$

Note: Initial voltages must all be equal!

$$\Rightarrow C_{eq} = C_1 + C_2 + C_3 + \ldots + C_n$$

Series Capacitors

Example: Find $v_{C1}(t)$.



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_n}$$

$$= 1 + \frac{1}{2} + \frac{1}{2} = 2$$

$$\Rightarrow C_{eq} = \frac{1}{2}\,\mu\text{F}$$

$$i = C\frac{dv}{dt}$$

$v_{C1}(0) = 10$

$v_S(t) = 3\sin 2t, \quad t \geq 0$

Series Capacitors

$$i_S(t) = C_{eq}\frac{dv_S(t)}{dt} = \begin{cases} C_{eq}\,6\cos 2t = 3\cos 2t \quad (\mu\text{A}) & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$v_{C1}(t) = \frac{1}{C_1}\int_0^t i_S(\tau)d\tau + v_{C1}(0) = \frac{1}{1}\int_0^t 3\cos 2\tau d\tau + 10$$

$$\boxed{v_{C1}(t) = 10 + \frac{3}{2}\sin 2t \quad (\text{V})}$$

Review: Passive Elements

- Three types of **ideal basic circuit elements**:
  - Resistors
  - Inductors
  - Capacitors
- Voltage-current, power relationships:

Resistors:

$$v_R = \pm R i_R \quad \Rightarrow i_R(t) = \pm \frac{1}{R} v_R \qquad p_R = \pm v_R i_R = +R i_R^2 = + \frac{v_R^2}{R}$$

Inductors:

$$v_L = \pm L \frac{di_L}{dt} \Rightarrow i_L(t) = \pm \frac{1}{L} \int_{t_0}^{t} v_L d\tau + i_L(t_0) \qquad p_L = \pm v_L i_L = +L i_L \frac{di_L}{dt}$$

Capacitors:

$$i_C = \pm C \frac{dv_C}{dt} \Rightarrow v_C(t) = \pm \frac{1}{C} \int_{t_0}^{t} i_C d\tau + v_C(t_0) \qquad p_C = \pm v_C i_C = +C v_C \frac{dv_C}{dt}$$

Review: Series and Parallel Equivalents

	Elements in series	Elements in parallel
Resistors	$R_{eq} = R_1 + R_2 + \ldots + R_n$	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$
Inductors	$L_{eq} = L_1 + L_2 + \ldots + L_n$	$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_n}$
Capacitors	$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_n}$	$C_{eq} = C_1 + C_2 + \ldots + C_n$

Chapter 7: Introduction (cont.)

We will first discuss circuits with only one energy-storing element: a single C or L.

- These circuits are called first-order circuits.
- This circuits are described by first-order differential equations.
- Sources + resistors + single C= RC circuit.
- Sources + resistors + single L= RL circuit.

Important remark:

a circuit with *N* energy-storing elements can have a max order *N*.

Why?

Chapter 7: Introduction (cont.)

In this chapter we will discuss the following:

Natural response: Response of a capacitive/inductive circuit to the initial conditions (there is no source):

- Initial voltage of capacitors
- Initial current of inductors

Step response: Response of a capacitive/inductive circuit to step inputs:

- Step/abrupt voltage changes (sources)
- Step/abrupt current changes (sources)

Step input: Sudden application of DC voltage or current or sudden change in the DC voltage or current input levels.

Input Response

- Step response is the response of a system to a step input.
- This is an special case of an input response.
- In system theory and identification we are interested in the response and behavior of systems to a number of well known and important excitations/inputs:
  - Ramp response
  - Impulse response
  - Sinusoidal response…

First Order RL and RC Circuits

Reminder:

RL= resistor + inductor + sources.

RC= resistor + capacitor + sources.

Remark:

If multiple inductors in an RL circuit or multiple capacitors in an RC circuit exist, in order to have a first-order circuit/behavior, they must be interconnected (series and/or parallel) such that they can be replaced by a single equivalent element.

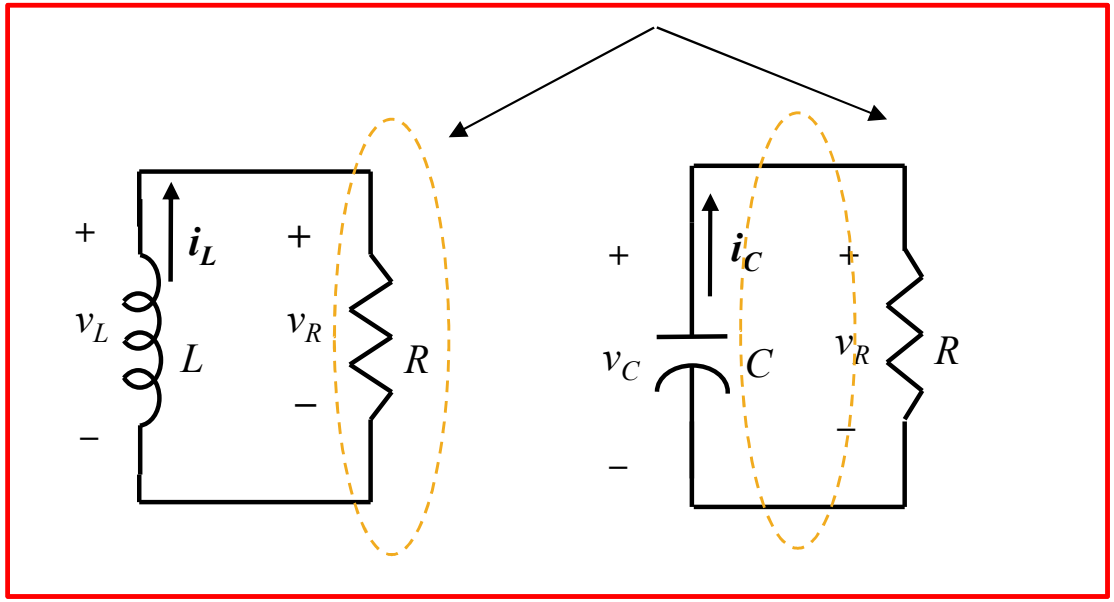
Note:

There are first-order circuits where the above is not possible. We do not analyze those at this point.

First Order Configurations

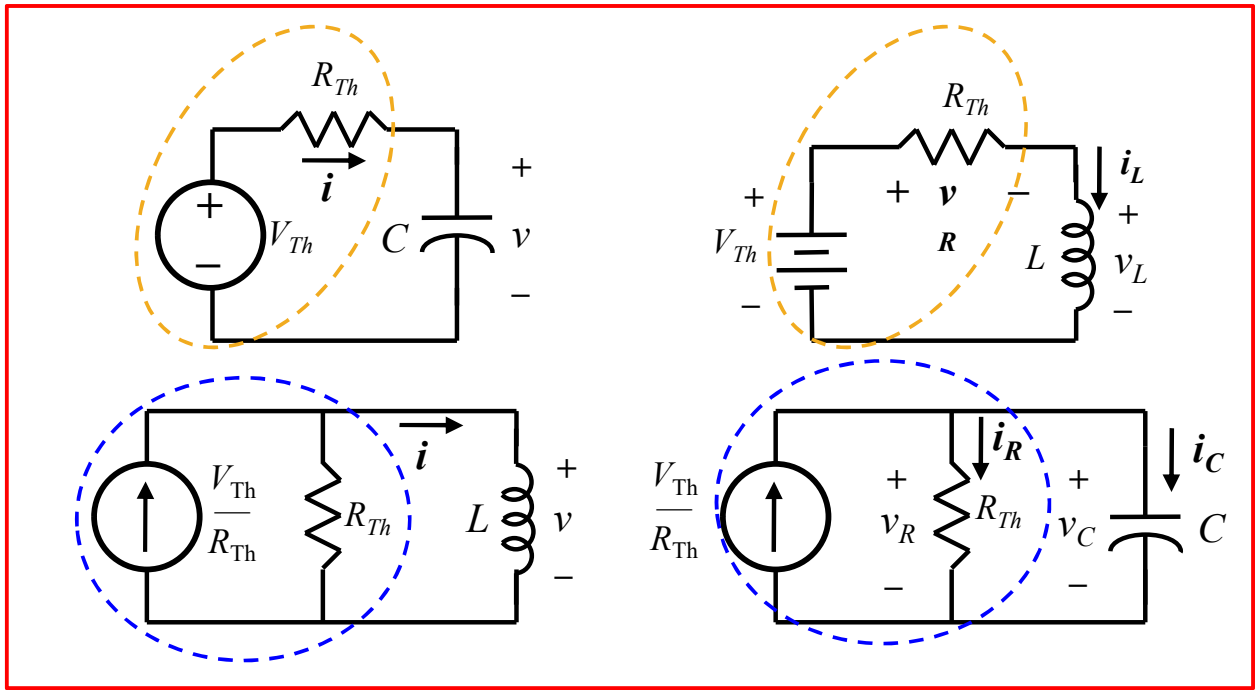
For natural response: The only sources of energy/work: initial conditions of C or L.

Equivalent circuit (resistance) from the eyes of the single energy-storing element



First Order Configurations (cont.)

For step response:



Equivalent circuit (**Thevenin** or **Norton**) from the eyes of the single energy-storing element

Are there any other possibilities? Are they any good?

RL and RC Responses

Natural response:

when there is no independent source in the circuit and the energy stored in the inductor or capacitor is suddenly released to a resistive network.

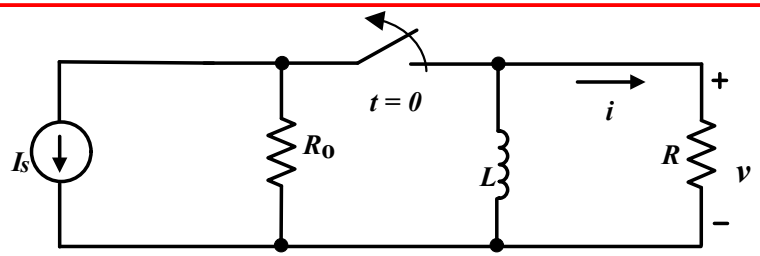


Figure 7.3 An *RL* circuit

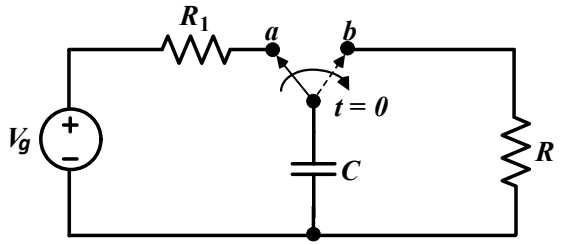


Figure 7.10 An *RC* circuit

Step response:

when a dc voltage or current source is suddenly connected to the circuit.

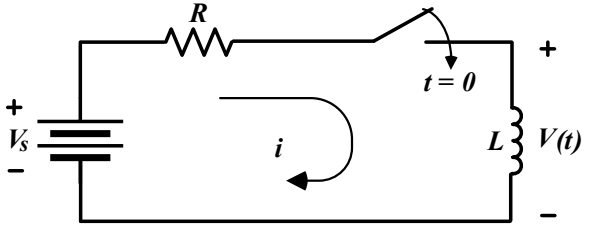
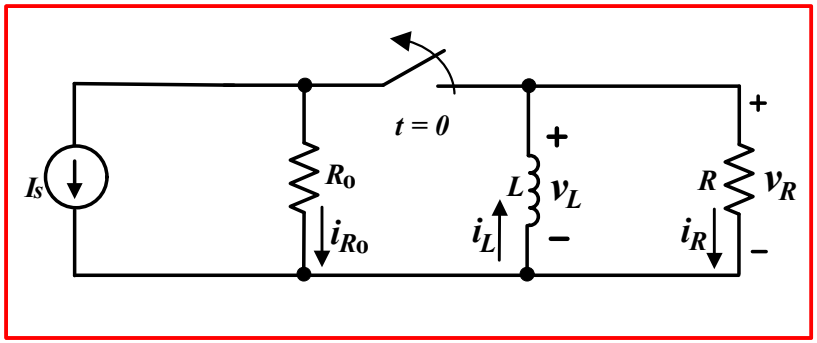


Figure 7.16 A circuit used to illustrate the step response of a first-order *RL* circuit.

Natural Response of an RL Circuit

Assumption: The switch has been in the closed position for a long time. At $t = 0$, the switch is opened.

Assumption: I_S is a constant (dc) current.



If the switch has been closed for a long time:
All currents and voltages have reached a constant value,
i_L = constant
$di_L/dt = 0$
$v_L = 0$ (i.e. inductor acts as a short circuit)

$\Rightarrow i_{Ro} = 0, \; i_R = 0$

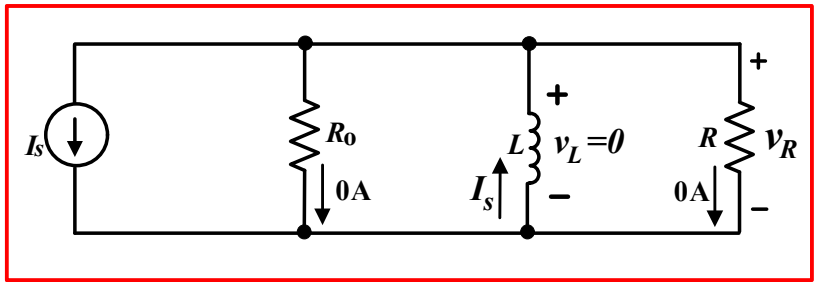
$i_L = i_{Ro} + i_R + I_S = I_S$ ➡ i.e., all the source current passes through the inductor L.

$\Rightarrow$ before switching (at $t = 0^-$):

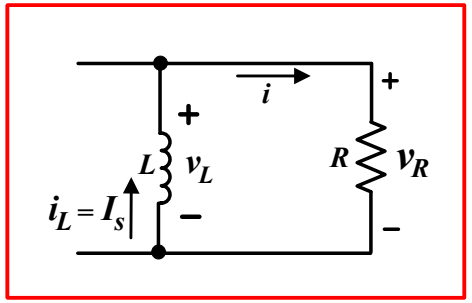
$\Rightarrow i_L(0^-) = I_S \,, \; v_R(0^-) = 0.$

Natural Response of an RL Circuit (cont.)

Before switching (t=0$^-$):



After Switching (at t=0$^+$):



Since current in an inductor can not change abruptly:

$$i_L(0^+) = i_L(0^-) = I_S = I_0 \quad \text{initial current}$$

$$\left.\begin{aligned} & i_L(t) = i_R(t) \equiv i(t) \\ & v_L(t) = -L\frac{di_L(t)}{dt} \\ & v_R(t) = Ri_R(t) \\ & v_R(t) = v_L(t) \end{aligned}\right\} \Rightarrow -L\frac{di(t)}{dt} = Ri(t) \quad \text{or} \quad \boxed{L\frac{di(t)}{dt} + Ri(t) = 0}$$

After Switching ($t \geq 0^+$)

1st order differential equation

Natural Response of an RL Circuit (cont.)

$$L\frac{di(t)}{dt}+Ri(t)=0$$

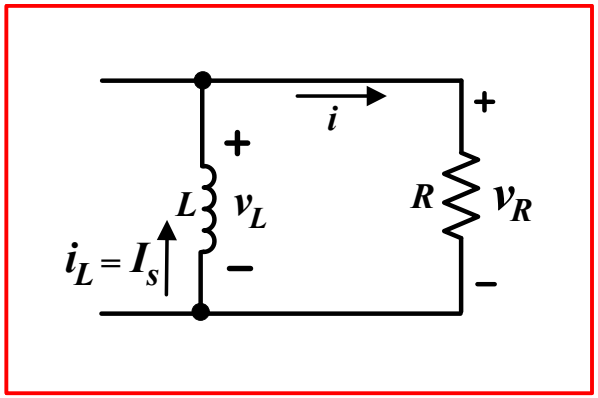
$$\Rightarrow \frac{di(t)}{i(t)}=-\frac{R}{L}dt \Rightarrow \int_{i(0^+)}^{i(t)}\frac{di'(t)}{i'(t)}=-\frac{R}{L}\int_{0^+}^{t}dt'$$

$$\Rightarrow \ln[i(t)]-\ln[i(0^+)]=\ln\frac{i(t)}{i(0^+)}=-\frac{R}{L}t \quad \boxed{\Rightarrow i(t)=i(0^+)\mathrm{e}^{-\frac{R}{L}t}}$$

$$\Rightarrow i_L(t)=\begin{cases} I_0 & t=0^- \\ I_0\mathrm{e}^{-\frac{R}{L}t} & t\geq 0^+ \end{cases} \quad , \quad v_R(t)=\begin{cases} 0 & t=0^- \\ RI_0\mathrm{e}^{-\frac{R}{L}t} & t\geq 0^+ \end{cases}$$

No jump in current!

Jump in voltage!



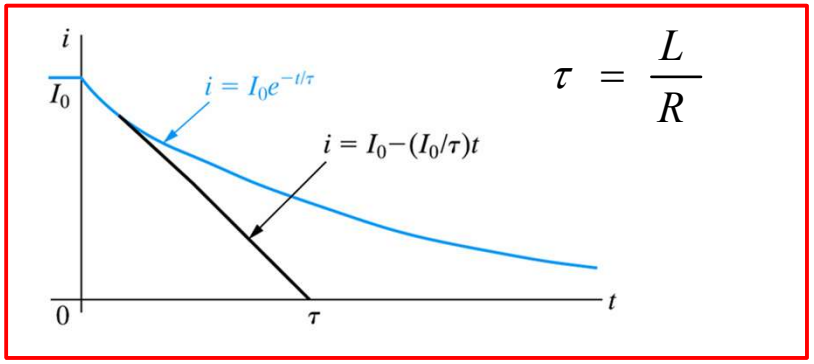
Natural Response of an RL Circuit (cont.)

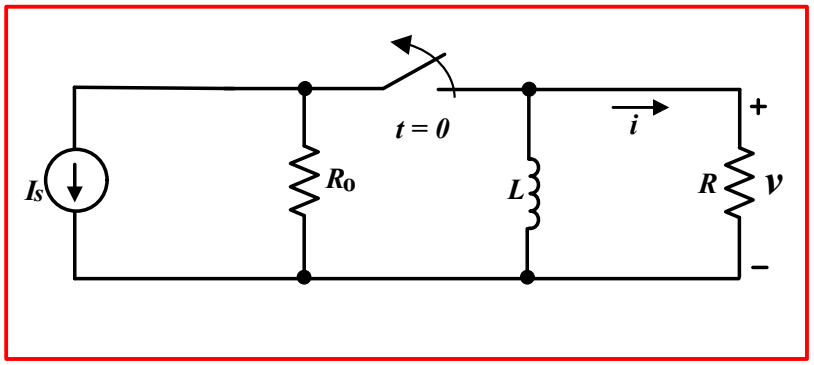
$$i_L(t) = \begin{cases} I_0 & t = 0^- \\ I_0 e^{-\frac{R}{L}t} & t \geq 0^+ \end{cases}$$

$$\lim_{t \to 0^+} i_L(t) = I_0 e^0 = I_0$$

$$\lim_{t \to \infty} i_L(t) = I_0 e^{-\infty} = 0$$

$$i_R(t) = \begin{cases} 0 & t = 0^- \\ I_0 e^{-\frac{R}{L}t} & t \geq 0^+ \end{cases} \quad \text{or} \quad i_R(t) = \begin{cases} 0 & t = 0^- \\ i_L(t) & t \geq 0^+ \end{cases}$$





Natural Response of an RL Circuit (cont.)

Power and energy:

$$p_R(t) = Ri_R^2(t) = \begin{cases} 0 & t = 0^- \\ RI_0^2 \mathrm{e}^{-\frac{2R}{L}t} & t \geq 0^+ \end{cases}$$

$$w_R(t) = w_R(0^-) + \int_{0^+}^{t} p_R(t')dt' = \int_{0^+}^{t} RI_0^2 \mathrm{e}^{-\frac{2R}{L}t'} dt'$$

$$\Rightarrow w_R(t) = \frac{RI_0^2}{\left(-\frac{2R}{L}\right)} \mathrm{e}^{-\frac{2R}{L}t}\Bigg|_{0^+}^{t} = -\frac{LI_0^2}{2}\left(\mathrm{e}^{-\frac{2R}{L}t} - 1\right) = \frac{LI_0^2}{2}\left(1 - \mathrm{e}^{-\frac{2R}{L}t}\right) \qquad t \geq 0$$

$$w_R(\infty) = \frac{LI_0^2}{2}$$

$$w_L(0^+) = \frac{1}{2}Li_L^2(0^+) = \frac{LI_0^2}{2}$$

$$\Rightarrow w_R(\infty) = w_L(0^+)$$

All the stored energy in L is finally dissipated in R.

Natural Response of an RL Circuit (cont.)

Time constant: Define $\tau = \text{time constant} \equiv \dfrac{L}{R}$.

$t = \tau \Rightarrow i_L(\tau) = I_0 e^{-1} = 0.367 I_0$ 63.3% drop in I_0

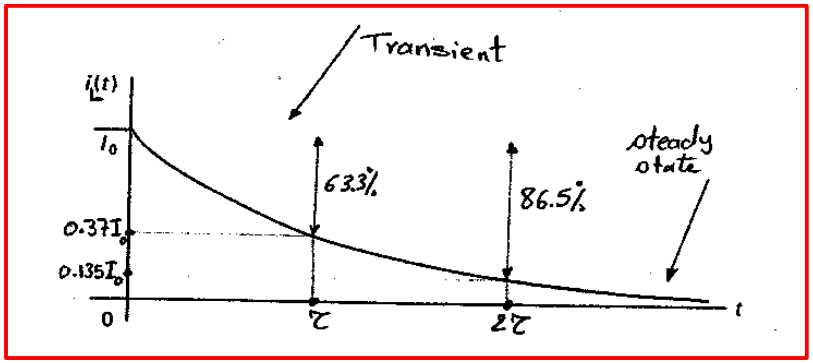
$t = 2\tau \Rightarrow i_L(2\tau) = I_0 e^{-2} = 0.135 I_0$ 86.5% drop in I_0

…

$t = 5\tau \Rightarrow i_L(5\tau) = I_0 e^{-5} = 0.0067 I_0$ 99.3% drop in I_0

$$i_L(t) = I_0 e^{-\frac{t}{\tau}} \quad t \geq 0^+$$

For $t \geq 5\tau$, steady-state (final) response is effectively achieved.

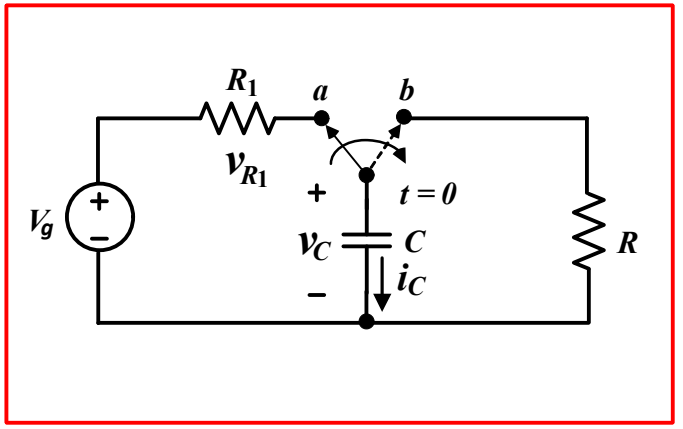


4τ to 5τ is effectively the long time or plus infinity time for 1st order systems!

Natural Response of an RC Circuit

Assumption: The switch has been in the closed position for a long time. At $t = 0$, the switch flips.

Assumption: V_g is a constant (dc) current



If the switch has been closed for a long time:

All currents and voltages have reached a constant value.

v_C = constant

$dv_C/dt = 0$

$i_C = 0$ (i.e. capacitor acts as an open circuit)

$\Rightarrow v_{R1} = 0$

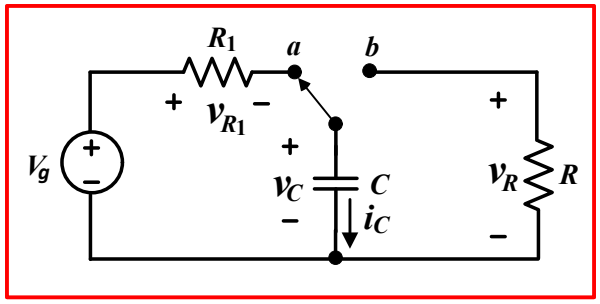
$v_C = V_g - v_{R1}$

$v_C = V_g$ → i.e. the source voltage is dumped on the capacitor C.

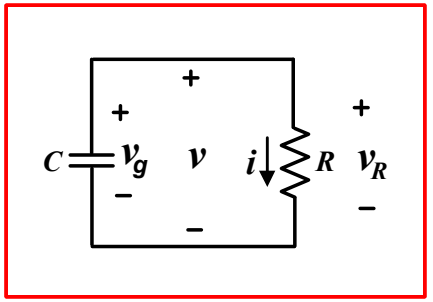
before switching ($t = 0^-$): $v_C(0^-) = V_g$

Natural Response of an RC Circuit (cont.)

Before switching (t=0⁻):



After Switching (t=0⁺):



Since voltage in a capacitor can not change abruptly:

$$v_C(0^+) = v_C(0^-) = V_g = V_0 \qquad \text{initial voltage}$$

$$\left.\begin{aligned} i_C(t) &= -C\frac{dv_C(t)}{dt} \\ i_R(t) &= \frac{v_R(t)}{R} \\ v_R(t) &= v_C(t) \equiv v(t) \\ i_C(t) &= i_R(t) \end{aligned}\right\} \Rightarrow -C\frac{dv(t)}{dt} = \frac{v(t)}{R}$$

After Switching ($t \geq 0^+$)

or $$C\frac{dv(t)}{dt} + \frac{v(t)}{R} = 0$$

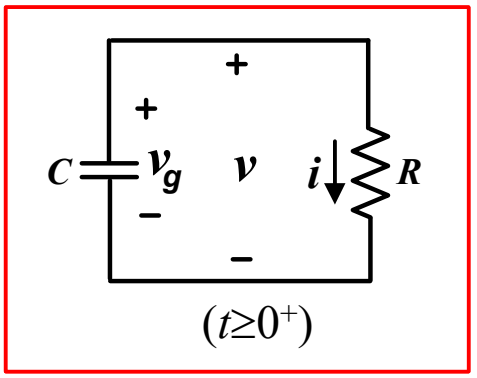
1st order differential equation

Recall: $$L\frac{di(t)}{dt} + Ri(t) = 0$$

Natural Response of an RC Circuit (cont.)

$$C\frac{dv(t)}{dt}+\frac{v(t)}{R}=0$$

$$\Rightarrow \frac{dv(t)}{v(t)}=-\frac{1}{RC}dt \Rightarrow \int_{v(0^+)}^{v(t)}\frac{dv'(t)}{v'(t)}=-\frac{1}{RC}\int_{0^+}^{t}dt'$$



$$\Rightarrow \ln v(t)-\ln v(0^+)=\ln\frac{v(t)}{v(0^+)}=-\frac{t}{RC} \quad \Rightarrow v(t)=v(0^+)e^{-\frac{t}{RC}}$$

$$\Rightarrow v_C(t)=\begin{cases}V_0 & t=0^-\\ V_0 e^{-t/RC} & t\geq 0^+\end{cases}, \quad i_C(t)=i_R(t)=\begin{cases}0 & t=0^-\\ \frac{V_0}{R}e^{-t/RC} & t\geq 0^+\end{cases}$$

No jump in voltage!

Jump in current!

Natural Response of an RC Circuit (cont.)

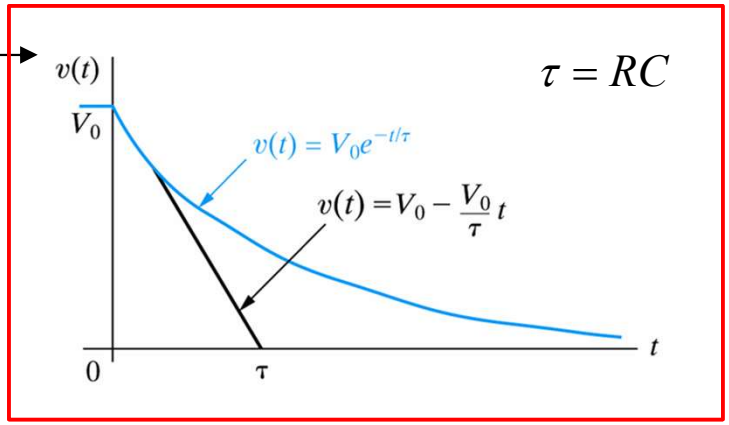
$$v_C(t) = \begin{cases} V_0 & t = 0^- \\ V_0 e^{-t/RC} & t \geq 0^+ \end{cases}$$

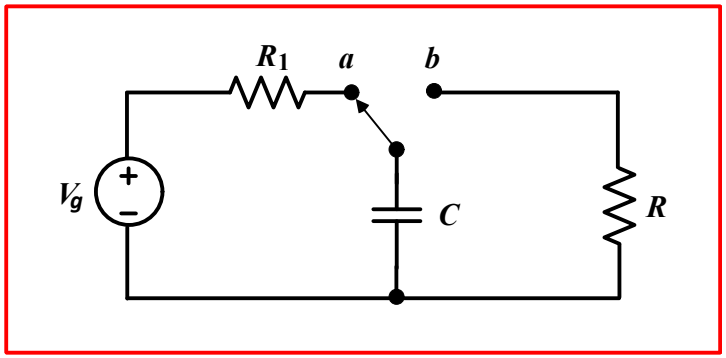
$$\lim_{t \to 0^+} v_C(t) = V_0 e^0 = V_0$$

$$\lim_{t \to \infty} v_C(t) = V_0 e^{-\infty} = 0$$

$$v_R(t) = \begin{cases} 0 & t = 0^- \\ V_0 e^{-t/RC} & t \geq 0^+ \end{cases}$$

$$i_R(t) = \begin{cases} 0 & t = 0^- \\ \dfrac{V_0}{R} e^{-t/RC} & t \geq 0^+ \end{cases}$$





Natural Response of an RC Circuit (cont.)

Power and energy:

$$p_c = \frac{dw_C}{dt} \Rightarrow dw_C = p_C dt$$

$$p_R(t) = \frac{v_R^2(t)}{R} = \begin{cases} 0 & t = 0^- \\ \dfrac{V_0^2}{R} e^{-\frac{2t}{RC}} & t \geq 0^+ \end{cases}$$

$$w_R(t) = w_R(0^-) + \int_{0^+}^{t} p_R(t')dt' = \int_{0^+}^{t} \frac{V_0^2}{R} e^{-\frac{2t'}{RC}} dt'$$

$$\Rightarrow w_R(t) = \frac{\frac{V_0^2}{R}}{\left(-\frac{2}{RC}\right)} e^{-\frac{2t}{RC}} \Bigg|_{0^+}^{t} = -\frac{1}{2} CV_0^2 \left(e^{-\frac{2t}{RC}} - 1 \right) = \frac{1}{2} V_0^2 \left(1 - e^{-\frac{2t}{RC}} \right) \qquad t \geq 0$$

$$w_R(\infty) = \frac{1}{2} CV_0^2$$

$$w_C(0^+) = \frac{1}{2} Cv_C^2(0^+) = \frac{1}{2} CV_0^2$$

$$\Rightarrow w_R(\infty) = w_C(0^+)$$

All the stored energy in C is finally dissipated in R.

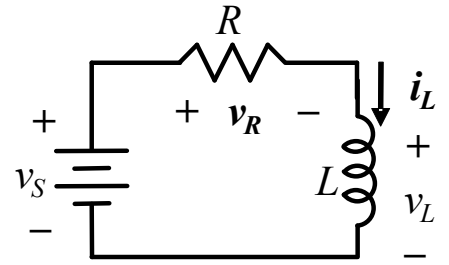
Step Response of an RL Circuit (cont.)

After switch closes:

Since current in an inductor can not change abruptly:

$$i_L(0^+) = i_L(0^-) = I_0$$

$$v_S = v_R + v_L = Ri_L(t) + L\frac{di_L(t)}{dt} \quad \text{(KVL)}$$



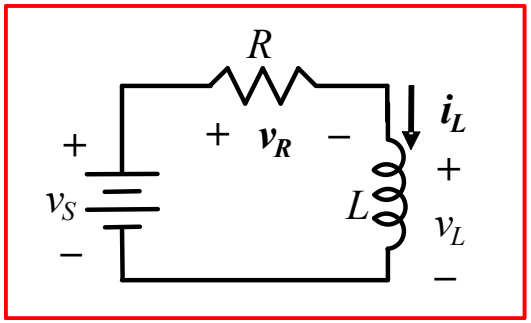
$$\Rightarrow i_L(t) = \frac{v_S}{R} + \left(I_0 - \frac{v_S}{R}\right)e^{-\frac{R}{L}t}, \quad t \geq 0^+$$

$$i_L(t) = i_L(\infty) + \left(i_L(0^+) - i_L(\infty)\right)e^{-\frac{t}{\tau}}$$

Step Response of an RL Circuit (cont.)

$$i_L(t) = \begin{cases} I_0 & t = 0^- \quad \text{(open)} \\ \dfrac{v_S}{R} + \left(I_0 - \dfrac{v_S}{R} \right) \mathrm{e}^{-\frac{R}{L}t} & t \geq 0^+ \quad \text{(closed)} \end{cases}$$

$$v_L(t) = L\frac{di_L(t)}{dt} \quad \Rightarrow v_L(t) = \begin{cases} 0 & t = 0^- \\ (v_S - RI_0)\,\mathrm{e}^{-\frac{R}{L}t} & t \geq 0^+ \end{cases}$$



In our case $i_L(0^-) = I_0 = 0$:

$$i_L(t) = \begin{cases} 0 & t = 0^- \\ \dfrac{v_S}{R}\left(1 - \mathrm{e}^{-\frac{R}{L}t}\right) & t \geq 0^+ \end{cases}, \quad v_L(t) = \begin{cases} 0 & t = 0^- \\ v_S\,\mathrm{e}^{-\frac{R}{L}t} & t \geq 0^+ \end{cases}$$

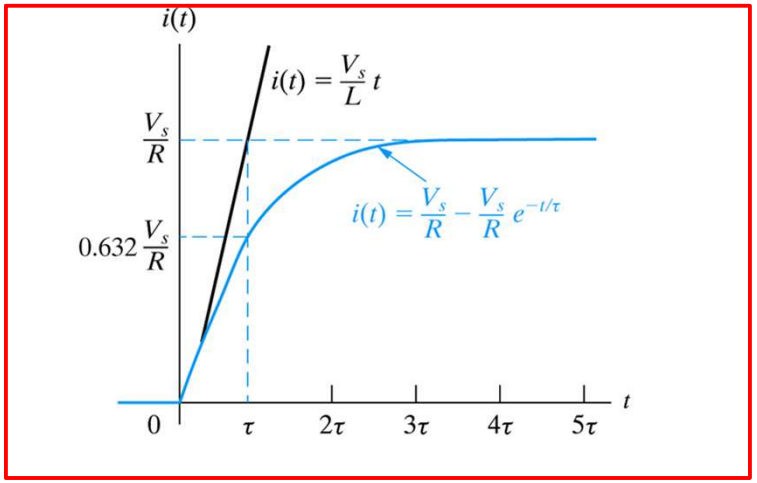
$$\text{and } i_R(t) = \begin{cases} 0 & t = 0^- \\ i_L(t) & t \geq 0^+ \end{cases}, \quad v_R(t) = \begin{cases} 0 & t = 0^- \\ Ri_L(t) = v_S + (RI_0 - v_S)\mathrm{e}^{-\frac{R}{L}t} & t \geq 0^+ \end{cases}$$

Step Response of an RL Circuit (cont.)

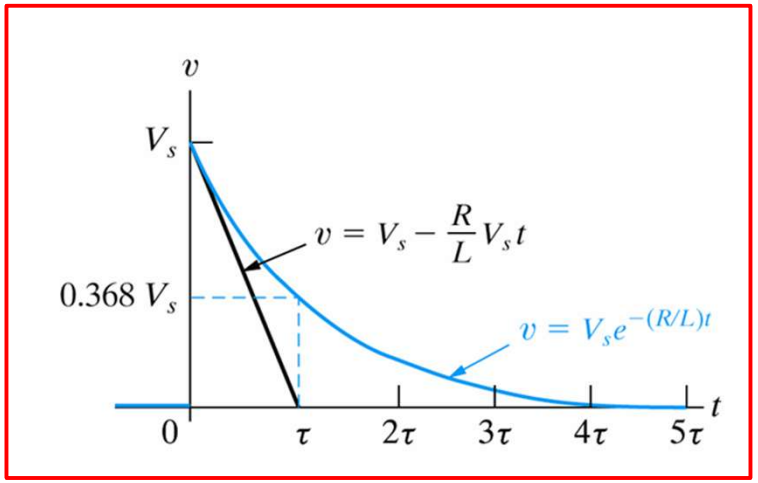
$$i_L(t) = \begin{cases} 0 & t = 0^- \\ \dfrac{v_S}{R}\left(1 - e^{-\frac{R}{L}t}\right) & t \geq 0^+ \end{cases}$$

$$, \quad v_L(t) = \begin{cases} 0 & t = 0^- \\ v_S e^{-\frac{R}{L}t} & t \geq 0^+ \end{cases}$$

$$\tau \equiv \frac{R}{L} = \text{time constant}$$



The step response of the RL circuit when $I_0 = 0$



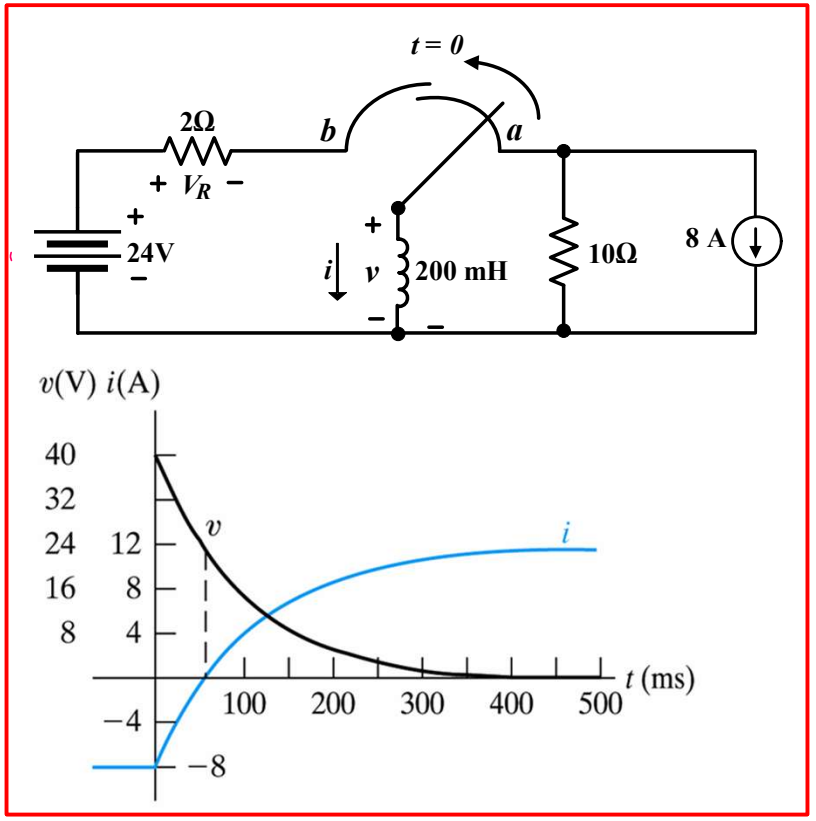
Inductor voltage vs. time

Example: Step Response of RL Circuit

The switch in the circuit shown has been in position "a" for a long time. At t = 0, the switch moves to "b".

a) Find $i(t)$ for $t \geq 0$.
b) What is the initial voltage across the inductor just after the switch has been moved to b?
c) Does the initial voltage make sense in terms of the circuit behavior?
d) How many milliseconds after the switch has been moved does the inductor voltage equal 24 V.
e) Plot $i(t)$ and $v(t)$ versus t.

Show up in Content Q&A on Wed. to see the solution.



Step Response of RC Circuits

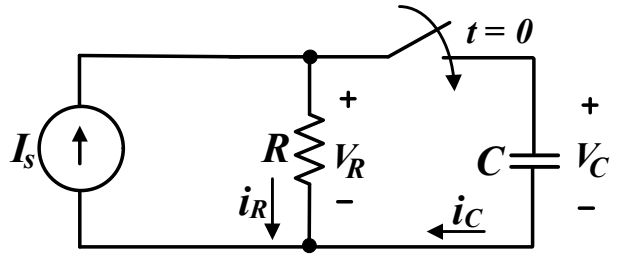
Assumption:

The switch has been in the open position for a long time:

The capacitor has an initial $Q_0 = CV_0$ on it (due to initial charge voltage).

At $t = 0$, the switch is closed.

Assumption: I_S is constant (dc current).



From assumption 1:

- $v_C(0^-) = V_0$
- $i_C(0^-) = 0$
- $i_R(0^-) = I_S$
- $v_R(0^-) = Ri_R(0^-) = RI_S$

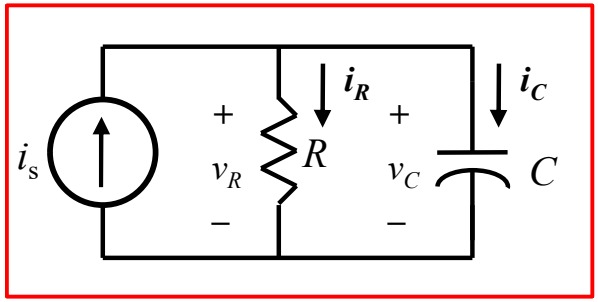
Step Response of RC Circuits (cont.)

After switch closes:

Since voltage in a capacitor can not change abruptly:

$$v_C(0^+) = v_C(0^-) = V_0$$

$$\Rightarrow \quad v_R(0^+) = V_0 \qquad \boxed{\Rightarrow i_R(0^+) = \frac{V_0}{R}}$$



$$i_C(0^+) = I_s - i_R(0^+)$$

$$\boxed{\Rightarrow i_C(0^+) = I_S - \frac{V_0}{R}}$$

$$i_C(t) + i_R(t) = I_S \quad \text{(KCL)}$$

$$\Rightarrow C\frac{dv_C(t)}{dt} + \frac{v_C(t)}{R} = I_S$$

$$\Rightarrow \frac{dv_C(t)}{dt} = \frac{1}{C}\left(I_S - \frac{v_C(t)}{R}\right)$$

$$\frac{dv_C(t)}{v_C(t) - RI_S} = \frac{-dt}{RC} \Rightarrow \int_{v_C(0^+)}^{v_C(t)} \frac{dv'_C(t)}{v'_C(t) - RI_S} = \frac{-1}{RC}\int_{0^+}^{t} dt'$$

$$\ln\left(\frac{v_C(t) - RI_S}{v_C(0^+) - RI_S}\right) = -\frac{t}{RC} \qquad \Rightarrow \frac{v_C(t) - RI_S}{v_C(0^+) - RI_S} = e^{-\frac{t}{RC}}$$

$$\boxed{\Rightarrow v_C(t) = RI_S + (V_0 - RI_S)e^{-\frac{t}{RC}}, \quad t \geq 0^+}$$

$$v_C(t) = v_C(\infty) + \left(v_C(0^+) - v_C(\infty)\right)e^{-\frac{t}{\tau}}$$

Step Response of RC Circuits (cont.)

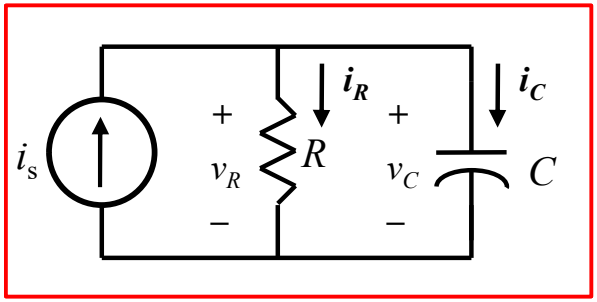
After switch closes:

Since voltage in a capacitor can not change abruptly:

$$v_C(0^+) = v_C(0^-) = V_0$$

$$\Rightarrow \quad v_R(0^+) = V_0$$

$$\Rightarrow i_R(0^+) = \frac{V_0}{R}$$



$$i_C(0^+) = I_s - i_R(0^+)$$

$$\Rightarrow i_C(0^+) = I_S - \frac{V_0}{R}$$

$$i_C(t) + i_R(t) = I_S \quad \text{(KCL)}$$

$$\Rightarrow C\frac{dv_C(t)}{dt} + \frac{v_C(t)}{R} = I_S$$

$$\Rightarrow v_C(t) = RI_S + (V_0 - RI_S)e^{-\frac{t}{RC}}, \quad t \geq 0^+$$

$$v_C(t) = v_C(\infty) + \left(v_C(0^+) - v_C(\infty)\right)e^{-\frac{t}{\tau}}$$

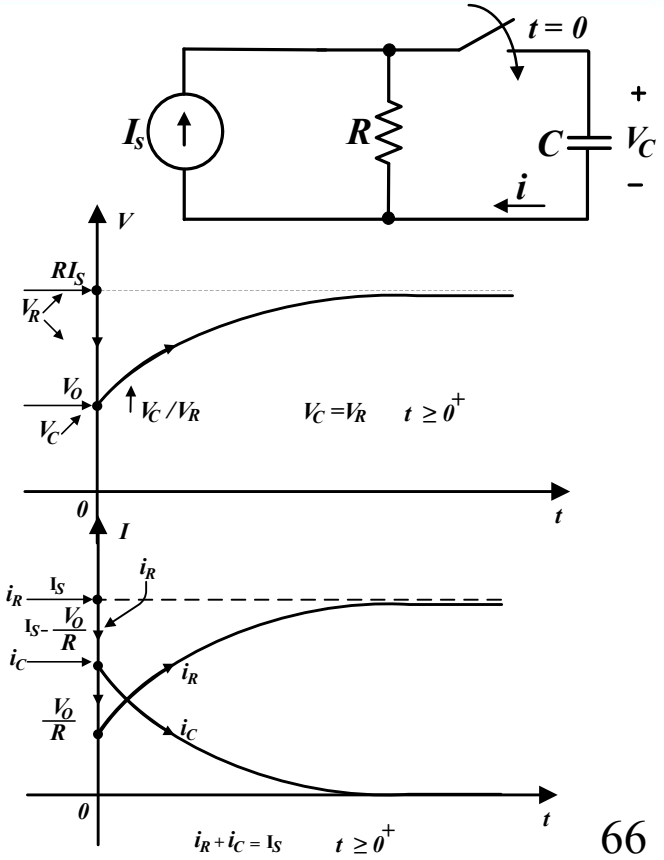
Step Response of RC Circuits (cont.)

$$v_C(t) = \begin{cases} V_0 & t = 0^- \quad \text{(open)} \\ RI_S + (V_0 - RI_S)e^{-\frac{t}{RC}} & t \geq 0^+ \quad \text{(closed)} \end{cases}$$

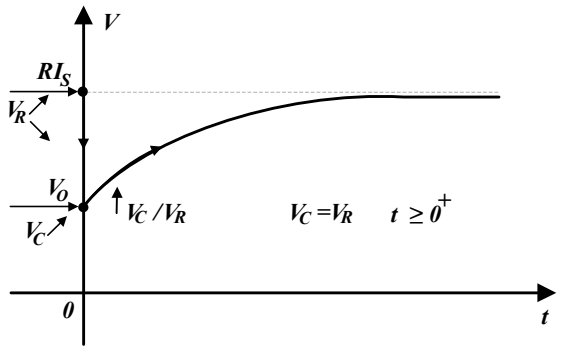
$$v_R(t) = \begin{cases} RI_S & t = 0^- \\ v_C(t) = RI_S + (V_0 - RI_S)e^{-\frac{t}{RC}} & t \geq 0^+ \end{cases}$$

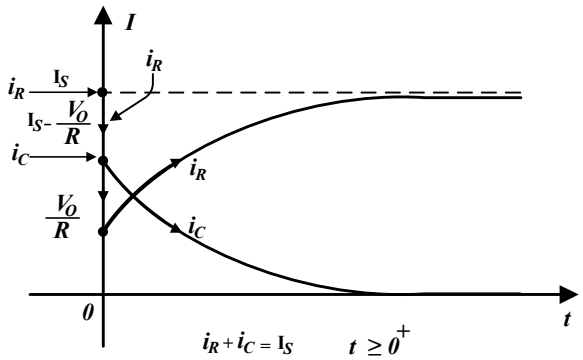
$$i_R(t) = \frac{v_R(t)}{R} \Rightarrow i_R(t) = \begin{cases} I_S & t = 0^- \\ I_S + \left(\frac{V_0}{R} - I_S\right)e^{-\frac{t}{RC}} & t \geq 0^+ \end{cases}$$

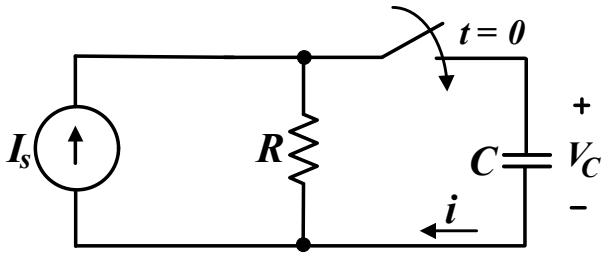
$$\Rightarrow i_C(t) = \begin{cases} 0 & t = 0^- \\ I_S - i_R(t) = -\left(\frac{V_0}{R} - I_S\right)e^{-\frac{t}{RC}} & t \geq 0^+ \end{cases}$$



Step Response of RC Circuits (cont.)







- $v_C(0^+) - v_C(0^-) = 0$ No jump
- $v_R(0^+) - v_R(0^-) = V_0 - RI_S$ Jump
- $i_R(0^+) - i_R(0^-) = V_0/R - I_S$ Jump
- $i_C(0^+) - i_C(0^-) = V_0/R - I_S$ Jump

$$\lim_{t\to\infty} v_C(t) = RI_S$$

$$\lim_{t\to\infty} i_C(t) = 0$$

$$\lim_{t\to\infty} i_R(t) = I_S$$

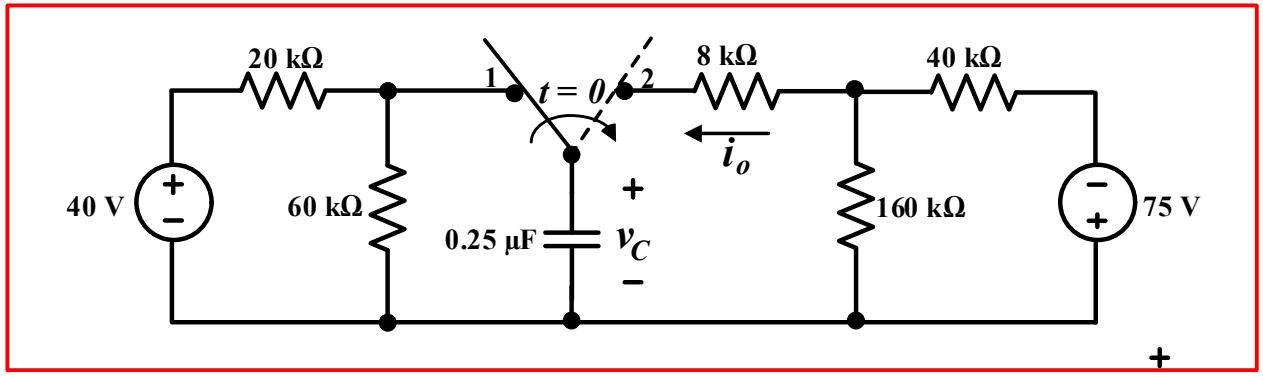
$$\lim_{t\to\infty} v_R(t) = RI_S$$

Step Response of RC Circuits (cont.)

Example:

The switch in the circuit shown has been in position 1 for a long time. At t = 0, the switch moves to position 2.

a)Find $v_C(t)$ for $t \geq 0$.
b)Find $i_0(t)$ for $t \geq 0^+$.



Show up in Content Q&A on Wed. to see the solution.